

TEST 1

Question 1:

Given:

$$\text{birth } (+1) \Rightarrow 0.4 \approx P(X_n = 1)$$

$$\text{Death } (-1) \Rightarrow 0.2 \approx P(X_n = -1)$$

$$\text{No change } (0) \Rightarrow 0.4 \{1 - 0.4 + 0.2\} \approx P(X_n = 0)$$

$$S_n = X_1 + X_2 + \dots + X_n$$

$$\{S_n; n = 0, 1, 2, 3, \dots\}$$

(i) What does X_n represent?

X_n represents the net number of births or deaths observed in the population after n years, starting from year 0.

(ii) State space: $S = \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\} = \{0, \pm 1, \pm 2, \dots\}$

(iii) Index set $n \geq 0$ represent time (in years) at which the observation about the population was made. $\{n \geq 0: \text{current year}, n \geq 1: \text{next year} \text{ and so on}\}$

(iv) $\{S_n, n \geq 0, 1, 2, \dots\}$. The future state X_{n+1} depends only on the present state X_n .

$$P_{ij} = P(X_{n+1} = j | X_n = i, X_{n-1} = X_{n-1}, \dots, X_0 = x_0)$$

$$\Rightarrow P(X_{n+1} = j | X_n = i) \text{ satisfies the Markov property.}$$

$$P_{ij} = P(S_{n+1} = j | S_n = i) = \begin{cases} 0.4, j = i+1 \\ 0.2, j = i-1 \\ 0.4, j = i \\ \text{or, otherwise} \end{cases}$$

(v) Yes! for any state i , we can reach any other state $j \in \mathbb{Z}$ by combination of $+1$ & -1 steps over time. $\forall i, j \in \mathbb{Z}, \exists n$ st

$$P_{ij} = 0.4 > 0$$

b) $E[X_2 | X_0 = 0]$

$$E[X_n] = E[X_0] + E[X_1] + \dots + E[X_{n-1}] = n \cdot E[X_1] \Rightarrow 2 \cdot 0.2 = 0.4$$

$$E[X_1] = 1(0.4) + 0(0.4) + (-1)(0.2) = 0.2$$

$$E[X_2 | X_0 = 0] = 2 \cdot 0.2 = 0.4$$

$$c) P(T_0=3 | X_0=0) = P(X_0=0, X_1 \neq 0, X_2 \neq 0 | X_0=0)$$

$$\Rightarrow P(X_1=1, X_2 \neq 0, X_3=0 | X_1=0) + P(X_1=-1, X_2 \neq 0, X_3=0 | X_1=0)$$

$$\Rightarrow [P(X_1=1, X_2=1, X_3=0 | X_0=0) + P(X_1=1, X_2=2, X_3=0 | X_0=0)] + [P(X_1=-1, X_2=1, X_3=0 | X_0=0) + P(X_1=-1, X_2=-2, X_3=0 | X_0=0)]$$

$$\Rightarrow [P(X_3=0 | X_2=1)P(X_2=1 | X_1=1)P(X_1=1 | X_0=0) + P(X_3=0 | X_2=2)P(X_2=2 | X_1=1)P(X_1=1 | X_0=0)] + [P(X_3=0 | X_2=1)P(X_2=1 | X_1=-1)P(X_1=-1 | X_0=0) + P(X_3=0 | X_2=-2)P(X_2=-2 | X_1=-1)P(X_1=-1 | X_0=0)]$$

$$[P(X_3=0 | X_2=1)P(X_2=1 | X_1=1)P(X_1=1 | X_0=0) + P(X_3=0 | X_2=2)P(X_2=2 | X_1=1)P(X_1=1 | X_0=0)] + [P(X_3=0 | X_2=1)P(X_2=1 | X_1=-1)P(X_1=-1 | X_0=0) + P(X_3=0 | X_2=-2)P(X_2=-2 | X_1=-1)P(X_1=-1 | X_0=0)]$$

$$\Rightarrow [P_{10} \cdot P_{11} \cdot P_{01} + P_{20} \cdot P_{12} \cdot P_{11}] + [P_{-10} \cdot P_{-11} \cdot P_{0-1} + P_{-20} \cdot P_{-12} \cdot P_{-11}]$$

$$\Rightarrow (0.2 \times 0.4 \times 0.4) + (0.4 \times 0.4 \times 0.2)$$

$$\Rightarrow 0.032 + 0.032$$

$$\Rightarrow 0.064$$

$$P(T_0=3 | X_0=0) = 0.064$$

Question 2

① Derive a Markov chain $\{X_n, n \geq 0\}$ that represents the number of deer in field B, and find the transition probabilities.

X_n = Number of deer in field B at day n

$$X_n = 0, 1, 2, \dots, 5$$

$$\{X_n, n = 0, 1, 2, 3, \dots\}$$

$$\text{State space } S = \{0, 1, 2, 3, 4, 5\}$$

$$p = 0.5, \text{ Total deer} = 5$$

$$X_{n+1} \sim \text{Bin}(5, 0.5)$$

Transition probabilities:

$$P_{ij} = P(X_{n+1} = j | X_n = i) = \binom{5}{j} (0.5)^5 \quad \forall i, j \in \{0, 1, 2, 3, 4, 5\}$$

$$P_{i,i+1} = 1 - \frac{i}{5} \quad i \in \{0, 1, 2, 3, 4, 5\}$$

$$P_{i,i-1} = \frac{i}{5}$$

② $\{X_n, n = 0, 1, 2, \dots\}$ = Irreducible & number of deer in field B start counting to any other number in one step. $\{P_{ij} > 0 \quad \forall i, j\}$

③ Since the chain is finite & irreducible, all states are positive recurrent.

④ Each day the number of deer in field B is binomially distributed with parameters $n = 5, p = 0.5$ and each new day are independent of the past.

$$E[X] = n \cdot p$$

$$E[X_n] = 5 \cdot 0.5$$

$$\text{Therefore } E[X_0 | X_0 = 3] = E[X_5] = 2.5$$

$$\textcircled{e} P(X_n = 3) = \binom{5}{3} \left(\frac{1}{2}\right)^5 \quad j = 0, 1, 2, \dots, 5 \text{ & } 5$$

$$\textcircled{f} P(X_0 = 3) = \binom{5}{3} (0.5)^5 = 10 \cdot \frac{1}{32} = \frac{10}{32} = \frac{5}{16} = 0.3125$$

$$\textcircled{g} E[T_3 | X_0 = 3] = \frac{1}{\pi_3}$$

$$\text{Stationary distribution } \pi_j = \binom{5}{j} (0.5)^5$$

$$\pi_3 = \binom{5}{3} (0.5)^5 = 10 \cdot \frac{1}{32} = \frac{10}{32}$$

$$E[T_3 | X_0 = 3] = \frac{1}{\pi_3} = \frac{32}{10} = 3.2$$

Question 3

Given:
Professions:

P: Professional

S: Skilled laborer

U: Unskilled laborer

X_n : Profession of a single randomly chosen son in n^{th} generation.

X_0 : Initial generation (man)

X_1 : 1st generation (son)

X_2 : 2nd generation (grandson)

X_n : n^{th} generation

$n = 0, 1, 2, \dots$

(i) X_n represents profession of a single son in the n^{th} generation

(ii) State Space $S = \{P, S, U\}$

(iii) Index set $n \geq 0$, n labels the generational number from the initial generation (man)

(iv) $\{X_n, n \geq 0\}$ is a Markov chain because it satisfies the properties associated with Markov:

$n = 0, 1, 2, 3, \dots$

State space $S = \{P, S, U\}$

$P_{ij} = P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j | X_n = i)$

(v) ① P P P S U

② S P S S P

③ U S U P P

(b) Transition Probabilities:

$$P_{ij} = \begin{pmatrix} P_P & P_S & P_U \\ P_S & P_S & P_U \\ P_U & P_S & P_U \end{pmatrix} = \begin{pmatrix} 0.8 & 0.1 & 0.1 \\ 0.2 & 0.6 & 0.2 \\ 0.25 & 0.25 & 0.5 \end{pmatrix}$$

$i, j \in \{P, S, U\}$

i row: Current generation's father's profession

j Column: Next generation son's profession

$$c) P(X_2 = p | X_0 = u) = P(X_2 = p, X_1 = i | X_0 = u)$$

$$\Rightarrow P(X_2 = p, X_1 = p | X_0 = u) + P(X_2 = p, X_1 = s | X_0 = u) + P(X_2 = p, X_1 = u | X_0 = u)$$

$$\Rightarrow P(X_2 = p | X_1 = p) P(X_1 = p | X_0 = u) + P(X_2 = p | X_1 = s) P(X_1 = s | X_0 = u) + P(X_2 = p | X_1 = u) P(X_1 = u | X_0 = u)$$

$$\Rightarrow P_{pp} P_{up} + P_{sp} P_{us} + P_{up} P_{uu}$$

$$\Rightarrow (0.8)(0.25) + (0.2)(0.25) + (0.25)(0.5)$$

$$\Rightarrow 0.2 + 0.05 + 0.125$$

$$\Rightarrow 0.375$$

$$d) Prob: X_4 = u, X_3 = p, X_2 = p, X_1 = s, X_0 = p$$

$$P(X_4 = u, X_3 = p, X_2 = p, X_1 = s, X_0 = p)$$

$$\Rightarrow P(X_4 = u | X_3 = p) P(X_3 = p | X_2 = p) P(X_2 = p | X_1 = s) P(X_1 = s | X_0 = p)$$

$$\Rightarrow P_{up} \cdot P_{pp} \cdot P_{ps} \cdot P_{sp}$$

$$\Rightarrow 0.25 \cdot 0.8 \cdot 0.1 \cdot 0.2$$

$$\Rightarrow 0.25 \cdot 0.8 \Rightarrow$$

$$\Rightarrow 0.004$$

Question 4:

$$X_n = q_1^n, \quad n \geq 0, 1, 2, 3, \dots$$

X_0 = Initial digit

X_1 = digit after 1st stage

X_2 = digit after 2nd stage

⋮

X_n = digit after nth stage

$S = \{0, 1\}$

(i) X_n represent the value of the transmitted digit at nth stage

(ii) state space S : $S = \{0, 1\}$

(iii) Index set $n \geq 0$ nth stage of transmission process

(iv) $\{X_n, n \geq 0\}$ is a Markov chain

$S = \{0, 1\}$

$$P(X_{n+1} = j | X_n = i, X_{n-1}, \dots) = P(X_{n+1} = j | X_n = i) \quad \forall i, j \in S$$

$$⑤ P(X_{n+1}=0 | X_n=0) = p_{00} = 0.35$$

$$P(X_{n+1}=0 | X_n=1) = p_{10} = 0.65$$

$$P(X_{n+1}=1 | X_n=0) = p_{01} = 0.65$$

$$P(X_{n+1}=1 | X_n=1) = p_{11} = 0.35$$

$$P_{ij} = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix} = \begin{pmatrix} 0.35 & 0.65 \\ 0.65 & 0.35 \end{pmatrix}$$

$$⑥: T \in \{1, 2, 3, \dots\}$$

$$① P(T=5) = P(X_1 \neq 1, X_2 \neq 1, X_3 \neq 1, X_4 \neq 1, X_5 = 1 | X_0 = 0)$$

$$\Rightarrow P(X_1=0, X_2=0, X_3=0, X_4=0, X_5=1 | X_0=0)$$

$$\Rightarrow P(X_3=1 | X_4=0) P(X_4=0 | X_5=0) P(X_3=0 | X_2=0) P(X_2=0 | X_1=0) P(X_1=0 | X_0=0)$$

$$\Rightarrow p_{00} p_{00} p_{00} p_{00} p_{01}$$

$$\Rightarrow q^4 p$$

$$\text{where } q = 0.35$$

$$p = 0.65$$

$$\Rightarrow (0.35)^4 \cdot (0.65)$$

$$\Rightarrow 0.00975$$

$$② P(T=n | X_0=0) = q^{n-1} \cdot p$$

$$\Rightarrow P(X_n=1 | X_{n-1}=0) P(X_{n-1}=0 | X_{n-2}=0) \dots P(X_1=0 | X_0=0)$$

$$\Rightarrow p_{01} p_{00} \dots p_{00}$$

$$\Rightarrow p_{01} (p_{00})^{n-1}$$

$$\Rightarrow p q^{n-1} = (0.65) (0.35)^{n-1}$$

$$④ P(T=n | X_0=1) = \frac{1}{p_{11}} \left\{ p_{10} = 0.65 \text{ or geometric distribution} \right\}$$

$$\therefore E[T_0 | X_0=1] = \frac{1}{0.65} = 1.54$$

Ques 11

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix} = \begin{bmatrix} 0.6 & 0.4 & 0.0 \\ 0.2 & 0.5 & 0.3 \\ 0.0 & 0.1 & 0.9 \end{bmatrix}$$

$$S = \{1, 2, 3\} \quad i, j \in S$$

1) $\{X_n, n \geq 0\}$ is Irreducible

for every state can be reached from every other state in finite number of steps

2) S is finite. The Markov chain is Irreducible.
Every state is recurrent.

Recurrent states = 1, 2, 3

Transient states = None.

3) $P(X_3=1 | X_0=1) = P(X_{n+3}=1 | X_n=1)$
 $\Rightarrow P^{(3)} = P^3$

Complete $P_{36}^2 = \begin{pmatrix} 0.6 & 0.4 & 0.0 \\ 0.2 & 0.5 & 0.3 \\ 0.0 & 0.1 & 0.9 \end{pmatrix} = P^2 = \begin{pmatrix} 0.44 & 0.44 & 0.12 \\ 0.22 & 0.36 & 0.42 \\ 0.02 & 0.14 & 0.84 \end{pmatrix} \Rightarrow P^3 = \begin{bmatrix} 0.352 & 0.408 & 0.240 \\ 0.204 & 0.310 & 0.486 \\ 0.040 & 0.112 & 0.198 \end{bmatrix}$

multiply Row 1 $\times P \Rightarrow P_{11}^3 = [(0.44 \times 0.6) + (0.44 \times 0.2) + (0.12 \times 0)]$

$P_{11}^3 \Rightarrow 0.352$

4) $E[X_3 | X_0=1] = 1 \cdot P_{11}^3 + 2 \cdot P_{12}^3 + 3 \cdot P_{13}^3$

$\Rightarrow 0.352 + 2(0.408) + 3(0.240)$

$\Rightarrow 1.888$

Question 6:

$$P_{ij} = \begin{cases} p & i \neq j \\ q & i = j \end{cases} \quad \text{where } p = 0.7, q = 0.3$$

$P_{ii} = 1 - P_{ii}$ {absorbing states}

$S = \{0, 1, 2, 3, 4, 5, 6\} \Rightarrow$ transient

$X_0 = 0$ & $X_6 = 6$ are absorbing states

$$T_j = \{n \geq 0: X_n = j\}$$

N_i = no. of visits to $j \geq \sum_{n=0}^{\infty} 1_{\{X_n=j\}}$

$$M_j^{(i)} = E[N_j | X_0 = i] = \sum_{n=0}^{\infty} P(X_n = j | X_0 = i)$$

To compute, we solve a system of equations for $M_j^{(i)} = i = 1, 2, \dots, 5$.

The recurrence for transient states is:

$$i \neq j \quad M_j^{(i)} = p \cdot M_j^{(i+1)} + q \cdot M_j^{(i-1)} \Rightarrow 0.7 M_j^{(i+1)} + 0.3 M_j^{(i-1)}$$

$$i = j \quad M_j^{(i)} = 1 + p \cdot M_j^{(i+1)} + q \cdot M_j^{(i-1)} \Rightarrow 1 + 0.7 M_j^{(i+1)} + 0.3 M_j^{(i-1)}$$

$$M_j^{(6)} = M_j^{(0)} = 0$$

② Expected visits to state 5, starting from 3.

Let $V_i = V_i^{(5)}$ if $i = 1$ to 5

$$V_1 = 0 + 0.7V_2 + 0.3(0)$$

$$V_2 = 0 + 0.7V_3 + 0.3V_1$$

$$V_3 = 0 + 0.7V_4 + 0.3V_2$$

$$V_4 = 0 + 0.7V_5 + 0.3V_3$$

$$V_5 = 1 + 0.7(0) + 0.3V_4$$

Solve these 5 eqs. simultaneously!

$$V_1 = 0.7V_2$$

$$\text{Sub } V_1 \text{ in } V_2: 0.7V_3 + 0.3V_1 = 0.7V_3 + 0.3(0.7V_2)$$

$$V_2 = 0.7V_3 + 0.21V_2 \Rightarrow V_2 - 0.21V_2 = 0.7V_3 \Rightarrow 0.79V_2 = 0.7V_3 \Rightarrow V_2 = \frac{0.7}{0.79} V_3 \approx 0.8861V_3$$

$$\therefore V_1 = 0.7V_2 = 0.7(0.8861V_3) \approx 0.6203V_3$$

$$\therefore V_3 = 0.7V_4 + 0.3V_2 = 0.7V_4 + 0.3(0.8861V_3) \Rightarrow V_3 - 0.2658V_3 = 0.7V_4 \Rightarrow$$

$$0.7342V_3 = 0.7V_4 \Rightarrow V_4 = \frac{0.7342}{0.7} V_3 \approx 1.049V_3$$

$$V_4 = 1.049V_3$$

$$V_5 = 1 + 0.3V_4 = 1 + 0.3(1.049V_3) \Rightarrow 1 + 0.3147V_3$$

$$V_3 = 0.7V_4 + 0.3V_2 \Rightarrow (0.7 \times 1.049 + 0.3 \times 0.8861) V_3$$

$$\Rightarrow (0.7343 + 0.2658) V_3 = 1.0001 V_3$$

We need to compute $V_5^{(s)}$

$$V_5 = 1 + 0.3V_4 = 1 + 0.3(1.049V_3) =$$

$$\Rightarrow 1 + 0.3147V_3$$

$$V_5^{(s)} \approx 1 + 0.3147 \cdot 1 = 1.3147$$

$$\text{Then } V_3 = 0.7V_4 + 0.3V_2 \Rightarrow V_3 = 0.7(1.049V_3) + 0.3(0.8861V_3) = 1.0001V_3$$

$$V_3 = 1$$

$$V_5 = 1 + 0.3(1.049)(1) = 1.3147$$

$$\therefore V_5^{(s)} = 1.3147$$

2. (1) Expected Number of Visits to state 1 from state 3

$$V_1 = 1 + 0.7V_2$$

$$V_2 = 0 + 0.7V_3 + 0.3V_1$$

$$V_3 = 0 + 0.7V_4 + 0.3V_2$$

$$V_4 = 0 + 0.7V_5 + 0.3V_3$$

$$V_5 = 0 + 0.3V_4$$

Solve from backward

$$V_5 = 0.3V_4$$

$$V_4 = 0.7V_5 + 0.3V_3 \Rightarrow 0.7(0.3V_4) + 0.3V_3 \Rightarrow 0.21V_4 + 0.3V_3 \Rightarrow V_4 - 0.21V_4 = 0.3V_3$$

$$\Rightarrow 0.79V_4 = 0.3V_3 \Rightarrow V_4 = \frac{0.3}{0.79} V_3 \approx 0.3797V_3$$

$$\text{Then: } V_5 = 0.3V_4 \approx 0.3(0.3797V_3) = 0.1139V_3$$

$$\text{Now: } V_3 = 0.7V_4 + 0.3V_2 = 0.7(0.3797V_3) + 0.3V_2 = 0.2658V_3 + 0.3V_2 \Rightarrow 0.7342V_3 = 0.3V_2$$

$$\Rightarrow V_2 = \frac{0.7342}{0.3} V_3 \approx 2.4473V_3$$

$$\Rightarrow V_1 = 1 + 0.7V_2 = 1 + 0.7(2.4473V_3) = 1 + 0.3(1 + 1.7131V_3) = 0.7V_3 + 0.3 + 0.5139V_3 = 1.2139V_3 + 0.3$$

$$\therefore V_2 \approx 2.4473V_3$$

$$\Rightarrow 2.4473V_3 = 1.2139V_3 + 0.3 \Rightarrow (2.4473 - 1.2139)V_3 = 0.3 \Rightarrow 1.2334V_3 = 0.3 \Rightarrow V_3 = \frac{0.3}{1.2334} \approx 0.2432$$

$$\therefore V_3^{(s)} \approx 0.2432$$